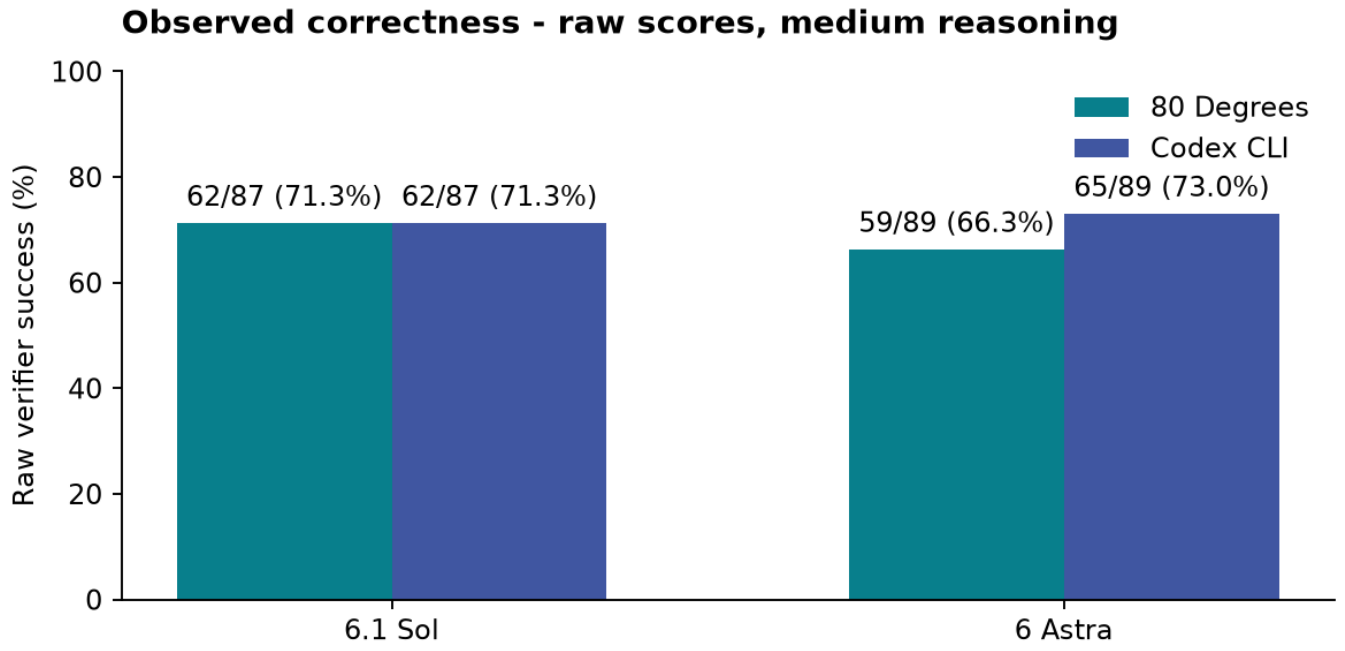
80 Degrees
Benchmark evidence

Investor assessment | Final conclusions on available evidence | 28 October 2026

Observed scorecard: Sol is a draw; Astra favors Codex on task success. 80 Degrees uses fewer total tokens and has lower Standard API-equivalent cost on complete raw pairs. These are completed measurements, not a cancelled study. Equal-correctness efficiency is a separate test that this study does not pass.

Question	Observed result and claim decision
A: More tasks solved	Sol: draw (62 vs 62). Astra: Codex ahead (65 vs 59). No 80 Degrees superiority claim.
B: Comparable correctness, fewer tokens	80 Degrees uses 23.8% / 27.5% fewer raw complete-pair tokens. The comparable-correctness requirement is not met.
B: Lower API-equivalent token cost	80 Degrees estimates are 1.6% / 12.1% lower on raw complete pairs. This is modeled cost, not subscription spending.



Raw zeroes include verifier failures; see the evidence-quality analysis.

Scope: 89 Terminal-Bench 2.0 tasks per model, medium reasoning, 80 Degrees native headless single-agent profile versus Codex CLI 0.163.0. 352 scored cells; four historical exclusions. This is first-party research, not independent certification.

Correctness and uncertainty

The paired comparison is the unit of analysis. Equal observed success is not evidence of equivalence. Negative point estimates are also not, by themselves, proof of inferiority.

Raw metric	6.1 Sol	6 Astra
Paired tasks	87	89
80 Degrees / Codex successes	62 / 62	59 / 65
Success difference (80D - Codex)	0.0 percentage points	-6.7 percentage points
80D-only / Codex-only solves	6 / 6	6 / 12
Exact McNemar p	1.0000	0.2379
Holm-adjusted p (two models)	1.0000	0.4758
Conservative 95% one-sided lower difference bound	-11.84 percentage points	-19.85 percentage points
Noninferiority requirement	Lower bound > -5 points	Lower bound > -5 points
Noninferiority established?	No	No

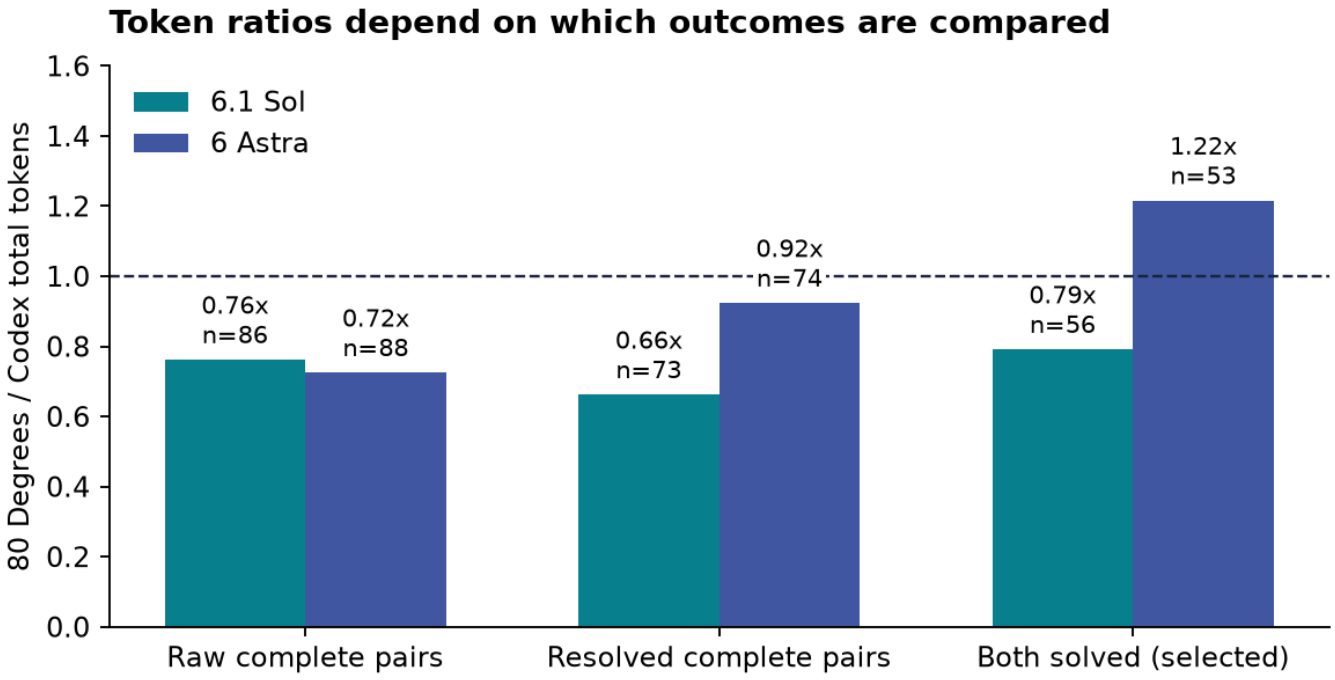
Evidence-quality sensitivity

Excluding pairs with an unavailable verifier leaves 74 Sol pairs (60 vs 61 successes) and 75 Astra pairs (58 vs 60). Their lower difference bounds remain below the -5-point margin: -13.58 and -16.09 points. Exclusion is a sensitivity analysis, not a replacement headline.

Across all 89 task slots, allowing unresolved outcomes and historical exclusions to take either value yields observed-difference ranges of **-14.61 to +14.61 points** for Sol and **-15.73 to +6.74 points** for Astra. These are missing-outcome bounds, not confidence intervals. Expanding unknowns to the mixed browser failures widens them further.

Decision: neither the predeclared statistical gate nor robustness to unresolved outcomes supports Claim A or the correctness component of Claim B.

Tokens: a qualified signal



Below 1 = fewer tokens. Complete telemetry only; subsets are descriptive.

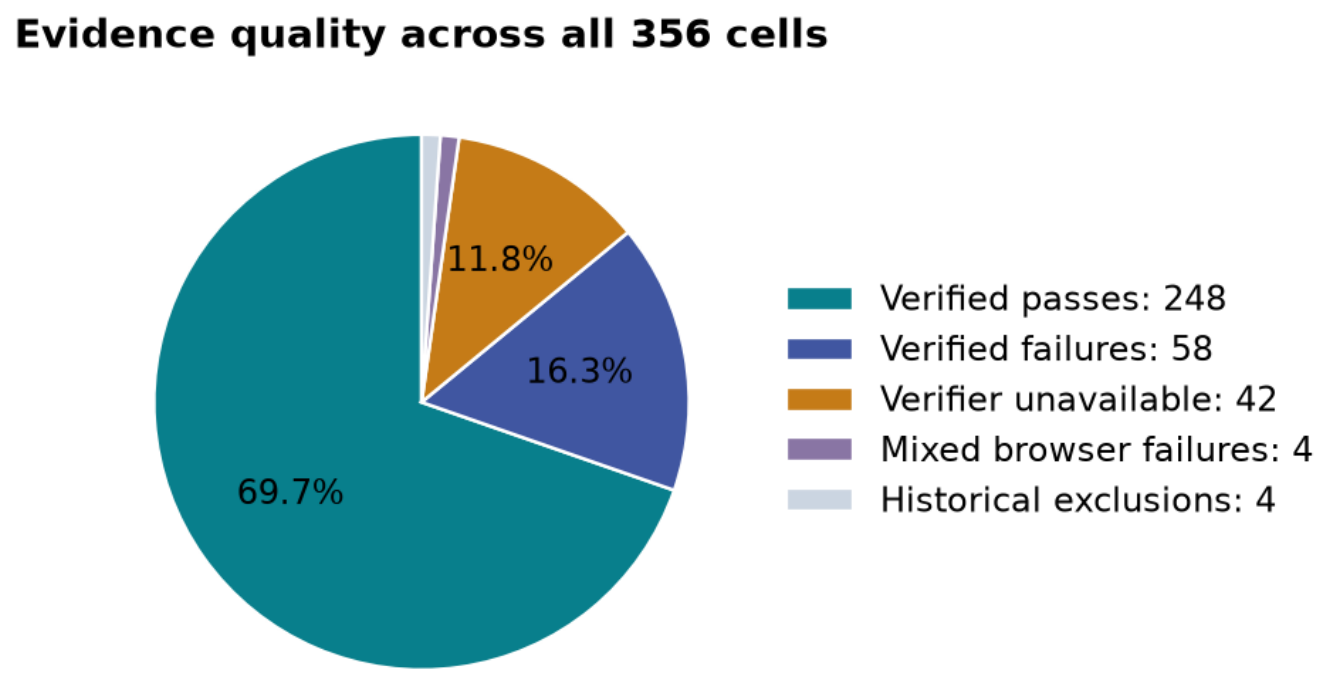
Complete raw pairs	6.1 Sol (86 pairs)	6 Astra (88 pairs)
80 Degrees input + output	29,679,726	25,281,642
Codex input + output	38,966,944	34,886,113
Observed reduction	23.8%	27.5%
Paired bootstrap central 95% ratio interval	0.563 to 1.047	0.488 to 1.035
One-sided 95% upper ratio bound	0.995	0.984

The correctness gate fails before token savings can substantiate Claim B. Even the two-sided token intervals include a ratio of 1. Both-solved subsets show 20.9% fewer tokens for Sol but **21.6% more for Astra**. These outcome-selected subsets are descriptive, not causal or equivalent-correctness tests.

Two native timeout trials have missing usage; complete pairs omit the matched Codex trials too. Cached tokens are included in input once; reasoning is included in output. No missing measurement is replaced with zero. Early unfinished exits can reduce tokens while reducing usefulness.

Cost: Standard API-equivalent estimates are now calculated from official rates and the observed cache mix; see the cost appendix. They favor 80 Degrees on raw complete pairs but favor Codex on both-solved pairs. They are not observed subscription charges.

What the evidence can support



42 raw zeroes have an unavailable verifier: 31 Python-download timeouts, nine network/package setup failures and two interrupted package-manager states. Four additional zeroes mix output assertions with browser dependency errors and remain failures in the primary raw view. The precise cause of every environment failure is not attributed to the provider or harness.

Recorded ledger disposition	Count
Scored cells	352
Historical reservations (four excluded cells)	10
Explicit provider-capacity failures	7
Local missing-runtime failure before model launch	1
Total recorded reservations	370

Defensible investor wording: "In a first-party, medium-reasoning Terminal-Bench 2.0 study, 80 Degrees matched Codex's raw Sol success count and had lower raw Astra success. On pairs with complete telemetry it used 23.8% and 27.5% fewer tokens, respectively. The study did not establish superior correctness, correctness noninferiority, or actual-dollar savings."

Do not shorten this to "same accuracy for less" or "outperforms Codex." Final analysis is complete for the available evidence; unresolved outcomes remain unresolved. No selective paid reruns were added to improve the result.

Reproducibility and limits

Component	Measured version / revision
80 Degrees source base	7a6e9e9acbd00eae1c0548a4c4f054b94c203faa
Measured Harbor adapter	770667538eb64aae2169b4006aa41ee781ae994c
Native engine checkout	028c1d7c7eb753421012bf96472a4ea040667f19
Codex CLI	0.163.0
Harbor	0.24.0 / 209a48a7424218828465c3c716b042f2b381dbed
Terminal-Bench 2.0	69671fbaac6d67a7ef0dfec016cc38a64ef7a77c

Native profile uses createAgentSession with read, bash, edit, write, grep and todo; no model fallback, delegation or automatic agent retry. Both arms use explicit medium reasoning and fresh sessions. The profile is not the full desktop experience. Apple Silicon host, linux/amd64 emulation; no claim about all hardware or agent versions.

Tasks were shuffled with seed 10262026, Sol ran before Astra and arm order alternated. The shared subscription and run order can affect capacity. Maximum three total launch reservations per cell included historical attempts. Scored failures were not retried as best-of-three; these are not pass@3 results. Four previous synthetic integration checks are outside the 370-reservation ledger.

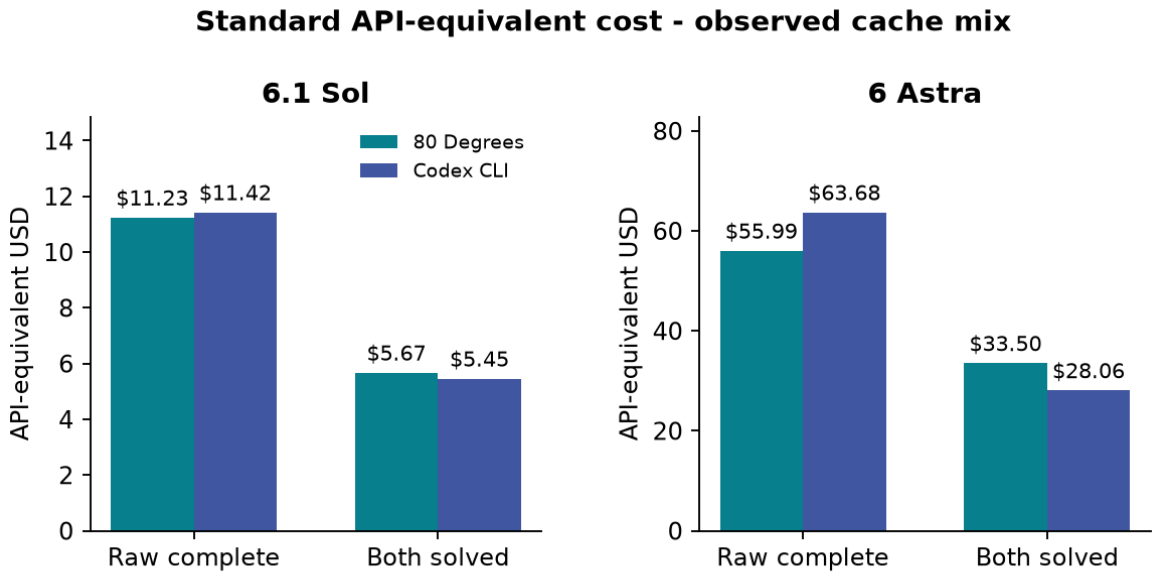
Exact paired McNemar with Holm adjustment across models; conservative paired win/loss bounds at alpha .025 per tail; predeclared 5-point noninferiority margin. Token ratios use 50,000 paired percentile-bootstrap samples, seed 10262026. Resampling characterizes task variation; it cannot remove bias from missing results, selected tasks or shared conditions.

The protocol followed earlier pilot observations, so it is not pristine preregistration. The parent source tree had a frontend lockfile modification outside the headless build; the actual engine checkout is recorded separately from the parent gitlink. Later portable-adapter commits were not used in this measurement.

Public evidence: github.com/80degrees-labs/80Degrees-Benchmarks, public/studies/october-2026. Includes task CSV/JSON, attempt accounting, classifications, hashes, analysis code, charts and both reports. Raw traces remain private; a reviewer must inspect authorized redacted artifacts to independently verify their contents. Hashes alone do not establish correctness. No SWE-bench or competitor-CLI conclusions are made.

API-equivalent cost scorecard

Official prices checked 28 October 2026. Standard text-token rates applied equally to both harnesses; no regional or priority premium. This answers a modeled-cost question, not actual subscription spending.



Estimated text-token cost, not subscription charges. Subsets have different coverage.

Measured subset	Sol: 80D / Codex	Astra: 80D / Codex
Raw complete pairs (86 / 88)	$11.23 / $11.42	$55.99 / $63.68
Observed cost reduction	1.6% lower	12.1% lower
Resolved complete pairs (73 / 74)	$8.34 / $8.82	$45.93 / $44.91
Both-solved pairs (56 / 53)	$5.67 / $5.45	$33.50 / $28.06
Both-solved cost difference	80D 3.9% higher	80D 19.4% higher

Formula: (uncached input x input rate + cache reads x cached rate + output x output rate) / 1,000,000. Zero cache writes were recorded. Rates per million: Sol $2 / $0.10 / $10; Astra $10 / $1 / $50, respectively. Reasoning and cached input are counted once.

No observed request exceeded the 272K-input long-context threshold. Two native aggregate/event residuals are below that threshold and priced from aggregate totals. Missing usage, retries, historical reservations, tool charges and subscription fees are excluded. Rate normalization does not prove what an actual API replay would cost.

Raw paired bootstrap 95% cost-ratio intervals: Sol [0.834, 1.159], Astra [0.689, 1.105]. Both include 1. A lower observed modeled total is a descriptive result, not proven equal-correctness cost superiority. Both-solved subsets are selected by outcomes.

Sources: developers.openai.com/api/docs/pricing; developers.openai.com/api/docs/models/gpt-6.1-sol; developers.openai.com/api/docs/models/gpt-6-astra. Reproduce with tooling/price_usage.py; numeric source-trace audit is in pricing-audit.json.